

General Principles of Antimicrobial Therapy AND Beta-Lactam Antibiotics, Other Cell Wall Inhibitors, and Membrane-Active Drugs, Part 1 and Part 2

Practice Quiz

Some items may have more than one answer.

The initial questions are intended to help students with pharmacology-related “must know” facts about pathogens.

To help you quickly find the answers for many of the questions, I have supplied page numbers of my Notes Handouts.

CHALLENGE: Cover the options and try to answer the questions without looking at the choices.

Goals of Practice Quizzes:

1. To help students identify what you know and what you understand by applying your knowledge to clinical case vignettes.
2. To help students get used to thinking through the “Board-style” case vignettes and questions. NYITYCOM, COMLEX and USMLE examine students’ medical knowledge using these types of questions.

HINT: Identify the information (clues) in the vignette that you need to answer the question. Don’t be distracted by the information that is not needed to answer the question.

If you wish to use them, all answers can be found by doing a “search” of my Notes Handout and/or the textbook chapters designated in the resources of each question. The handout and

Drugs and Bugs – Mechanisms, Mechanisms, Mechanisms

The clinical use of antibiotics is to eradicate pathogenic microorganisms.

To select the right drug, the clinician must know the microbiology of the pathogens.

Students, you need to know the bugs (bacteria).

And, of course, the clinically relevant pharmacokinetic properties and PK-PD profile is necessary knowledge for optimization of dosing and patient care.

Textbook readings are not a required, but you may find one or the other to be VERY helpful

- **Pharmacology bridges physiology, pathophysiology, and clinical practice.**

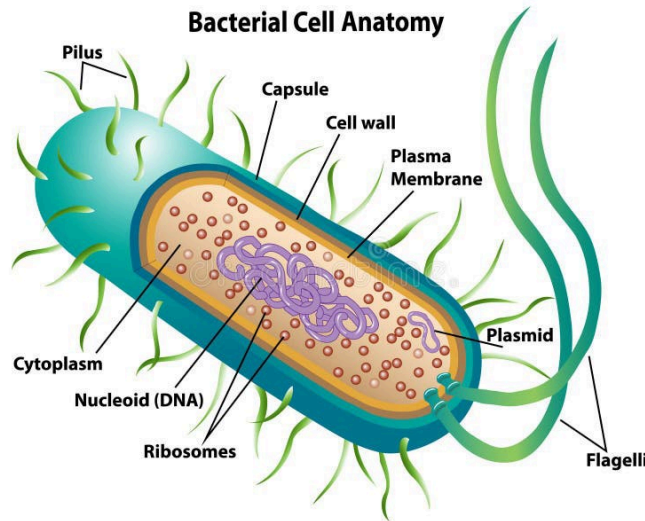
THE ANSWERS WILL BE POSTED NOV 21 BEFORE THE PRECOMP EXAM.

Practice points and some microbiology points are included in the answers document to provide *context and clinical relevance*. Students are not expected to memorize them for this course. My hope is that they will help you understand the pharmacology by conceptualizing it.

When
will the
answers
be
posted?

Learning Tip:

ELABORATION Finding additional layers of meaning by asking yourself “why” and “how” questions improves your *mastery* of new material and multiplies the mental cues available to you for later recall and application.



<https://www.dreamstime.com/illustration/bacterial-capsule.html>

PREFACE: Bacteria review relevant to antibiotic pharmacology

Relevance

Knowing these very basic concepts will help you understand which drugs are used for which bacterial infections and why (mechanisms of action and resistance related to therapeutic uses), even if you haven't had microbiology yet.

- What is the term for a microscopic single-celled organism that does not have specialized organelles or a nucleus with a nuclear membrane?
- What is the term for bacteria having a spherical or oval shape?
- What is the term for bacteria having a straight or curved shaped?
- What bacteria have a rigid, thick peptidoglycan wall, teichoic acids covalently linked to N-acetylmuramic acid (NAM) monomers, and lack an outer membrane?
- How are these bacteria categorized in terms of Gram stain?
- What bacteria have a lipopolysaccharide outer layer, a thin peptidoglycan layer, and a periplasmic space?
- What is the name of the proteins that allow hydrophilic solutes to diffuse across the outer layer of these bacteria?
- How are these bacteria categorized in terms of Gram stain?
- What bacteria are spiral shaped and either rigid or flexible and undulating?



<https://www.britannica.com/science/>

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j. Some bacteria have a polysaccharide layer that protects them from the immune system and contributes to virulence. What is this layer called?

k. Many bacteria contain small, usually circular, covalently closed, double-stranded DNA separate from the chromosome (“extra-chromosomal”). What are these structures called?

l. **Why are these extrachromosomal structures important in bacteriology and crucial in pharmacology?**

m. Some bacteria grow and maintain cellular functions if oxygen is available (metabolism by respiration) and in the absence of oxygen by fermentation. What is the term used to classify microorganisms able to use both metabolic pathways?

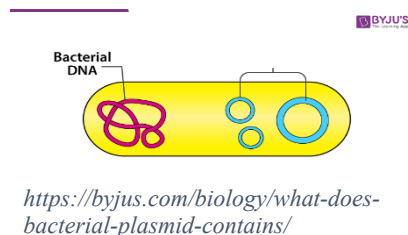
n. A large and diverse family of bacteria fitting this metabolism classification (above) are colonizers of the lower GI tract microbiome. What is the name of this large bacterial family (Handout p2)

o. Why is knowledge of this order of bacterial crucially important in clinical practice?

p. Some bacteria can exist in a small, dehydrated, metabolically quiescent (quiet) state in response to limited nutrients or adverse conditions (survival morphotype). Then, with appropriate stimulation, germinate into single vegetative cells. What are the metabolically quiescent morphotypes called?

q. Which organism of the type described in item p. is an anaerobic gram-positive bacillus that is resistant to nearly all antibiotics and causes infection as a result of antibiotic therapy?

r. Some bacteria cause community-acquired pneumonia caused by organisms not visualized on a Gram stain or cultured using traditional techniques AND they are intrinsically resistant to beta-lactam antibiotics. What are they?



YOU CAN FIND THE ANSWERS IN the Notes Handout, Microbiology textbook, and ScholarRx. Please use the reference you like best.

1. Thinking about the structural feature of the beta-lactam antibiotics necessary for its activity, explain how a particular enzyme expressed by bacteria, which is especially prevalent in gram-negative bacteria, inactivates the drugs.

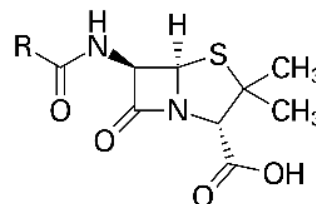
Short phrase

Hints: Relate it to the structure activity relationship.

Beta-Lactams Notes Handout, pages 6,8, and 10.

Brick: General Microbiology > Antimicrobial Agents > Penicillin

https://usmle-rx.scholarrx.com/rx-bricks/brick/CP_MIC0013



Learning Tip: GENERATION

The mind becomes more receptive to learning when you attempt to answer a question or solve a problem before being shown the answer or solution.

2. Beta-lactam antibiotics interfere with the actions of an essential microbial biochemical component (the target).

i) Where is this target located in the microbes?

- A. Cell wall
- B. Cytoplasmic membrane
- C. DNA
- D. Ribosome

ii) What is the ultimate killing effect of this antibacterial action?

- A. Apoptosis
- B. Autophagy
- C. Depolarization
- D. Osmotic rupture
- E. Oxidative stress

3. A medical student shadowing a family physician is asked to explain what is meant by selectivity when used in reference to antimicrobial therapy. The student correctly responds:

- A. The drug harms microbial cells but does not damage host cells.
- B. The drug targets only one type of microbial “receptor”.
- C. The proportion of actual positives that are not overlooked (few false negatives).
- D. The proportion of actual negatives that are correctly identified (few false positives).

4. An 8-year-old child is given oral penicillin VK every 8 hours for 10 days to be taken on an empty stomach with water for the treatment of streptococcal pharyngitis. Thinking about the class properties of the penicillins subclass, what is the antibiotic’s route of elimination?

Short answer

Learning Tip: APPLICATION

Applying your knowledge to practice – case vignettes at this point – help you assess your understanding.

5 K.C. is a 42-year-old healthy man presenting with a large, painful “boil” on the back of his neck. It started as small, tender bumps “a few days ago” and has gotten progressively worse. He denies fever or chills. No known drug allergies (NKDA).

The involved area is violaceous and indurated with multiple pustules draining externally around multiple hair follicles. Gram stain shows gram-positive spherical structures in grape-like irregular clusters.

He is given a diagnosis of carbuncles. The abscesses are drained; a sample of the exudate is sent for culture and susceptibility testing.

An e-prescription is submitted for an oral antibiotic with excellent activity against a penicillinase-producing species of the infective organism. He is directed to take it four times a day for 5 days one hour before eating or two hours afterward with at least four ounces of water and that it might cause nausea or heartburn, which is temporary and not harmful.

What antibiotic has been prescribed to this patient for definitive therapy?

- A. Amoxicillin
- B. Cefdinir
- C. Dicloxacillin
- D. Penicillin VK
- E. Vancomycin

ii) If K.C. reported that he got a rash from penicillin when he was a teenager, what drug would be an effective alternative?

Short answer

Definitions

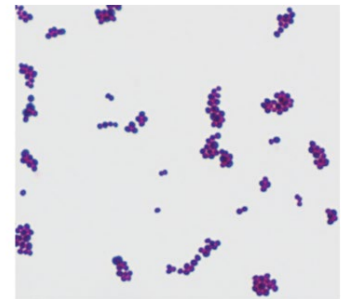
Carbuncle, a superficial skin infection: Multiple abscesses developing in hair follicles, sebaceous glands or sweat glands that form a connected area of infection under the skin. Common areas are the back of the neck, face, axillae, and buttocks. It is caused by a commensal organism of skin flora, which also colonizes anterior nares.

Hint: The Gram stain findings are hallmark microscopic features of *S. aureus*.



Source: Goldsmith LA, Katz SI, Gilchrist BA, Paller AS, Leffell DJ, Wolff K: *Fitzpatrick's Dermatology in General Medicine, 8th Edition*: www.accessmedicine.com

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Source: S. Riedel, J.A. Hadden, S. Miller, S.A. Morse, T.A. Mattner, B. Detrick, T.G. Mitchell, J.A. Sakanari, P. Hoter, R. Mejia Jawetz, Melnick, & Adelsberg's Medical Microbiology, 28e Copyright © McGraw-Hill Education. All rights reserved.

Gram-stain of *Staphylococcus aureus* showing Gram-positive cocci in pairs, tetrads, and clusters. Original magnification $\times 1000$. (Courtesy of L. Ching.)



Citation: Chapter 13 The Staphylococci. Riedel S, Hadden JA, Miller S, Morse SA, Mattner TA, Detrick B, Mitchell TG, Sakanari JA, Hoter P, Mejia Jawetz R, Melnick B, Adelsberg's Medical Microbiology, 28e. 2015. Available at: <https://accessmedicine.mhmedical.com/content.aspx?sectionid=517779581&bookid=26258&resultid=26258> Accessed: October 26, 2020

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6. A 27-year-old woman (G1P0) at 36 weeks gestation presents to the ED with fever, chills, myalgia, back pain, nausea, vomiting, and diarrhea. She reports that she was at a birthday party 10 days ago where she ate cheeses, deli meats, raw vegetable salad, and chocolate ice cream cake. She has had a healthy pregnancy to date. NKDA. Blood cultures are obtained, and the patient is admitted for intravenous antibiotic treatment of presumptive listeriosis.

The selected beta-lactam antibiotic, which was developed with an amino side chain to improve bacterial penetration and binding affinity, is highly active against the multidrug resistant (MDR), gram-positive rod *Listeria monocytogenes*. This drug is one of only a few that is effective against this organism.

Definition G1P0: Pregnant for the first time and has not yet delivered

G: gravida; P: para

i) Which one of the following best fits the description of the selected antibiotic?

- A. Ampicillin
- B. Aztreonam
- C. Cefazolin
- D. Meropenem
- E. Nafcillin
- F. Vancomycin

ii) What is the structure-activity relationship of the amino side chain?

Hint: Beta-lactams Notes Handout p27 and p29.

7. An adult patient with right-sided endocarditis caused by VRE is given antibiotic therapy with a drug active against gram-positive bacteria only. It apparently kills bacteria by causing leakage of potassium and loss of membrane potential, resulting in rapid death by interrupting DNA, RNA, and protein synthesis. To mitigate an adverse effect associated with the drug, the patient's plasma creatine kinase (CK) levels are monitored.

i) What does VRE stand for?

ii) What is the name of the drug?

iii) What drug-induced adverse effect is signaled by the elevated plasma CK levels?

- A. *C. difficile* superinfection
- B. Hepatocyte damage
- C. Myocardial infarction
- D. Myopathy
- E. Renal dysfunction

Hint: Beta-lactams Notes Handout p61.

8. A 40-year-old female hematopoietic stem cell transplant recipient receiving bone marrow-ablative conditioning with chemotherapy is treated for febrile neutropenia with empiric antimicrobial therapy that includes an antibiotic effective against a broad range of gram-positive and gram-negative bacteria, including *Pseudomonas aeruginosa*. The selected drug is co-formulated with cilastatin. Patient monitoring includes signs and symptoms of seizures.

What class is this drug in?

- A. Carbapenems
- B. Cephalosporins
- C. Cyclic lipopeptides
- D. Glycopeptides
- E. Monobactams
- F. Penicillins

Hint: Beta-lactams Notes Handout p51-52

9. What is the function of cilastatin in the antibiotic co-formulation being given to the hematopoietic stem cell transplant recipient above?

Short phrase

10. AK is a 76-year-old male hospitalized for the treatment of hematogenous osteomyelitis resulting from hip replacement surgery. The area is surgically debrided and tissue samples are sent for culture and sensitivity testing. The infective organism is identified as MRSA. Therapy with a beta-lactam antibiotic is avoided.

What is this organism's mechanism of resistance to the entire beta-lactams group?

(exception: Ceftaroline and ceftobiprole, which are not currently used for treatment of osteomyelitis.)

- A. Abnormally thick cell wall
- B. D-alanyl-D-lactate expression
- C. PBP with low affinity for beta-lactams
- D. Porins impenetrable to beta-lactams
- E. TEM1 β -lactamases

Hint: Beta-lactams Notes Handout p9.

11. What anti-MRSA drug from among the antibiotics studied in this block should be included in AK's empiric treatment of osteomyelitis?

One word answer

Learning Tips:

RETRIEVAL PRACTICE

Quizzing is a reliable measure of what you have learned and what you have not mastered.

SPACED PRACTICE

12. A few minutes into the second infusion of the anti-MRSA drug, AK complains of itching. He denies dyspnea and pain. Flushing and erythema are observed on his neck, face, and trunk. His vital signs are within normal limits. It is noted that the rate of infusion on the infusion pump is set too high.

What is the most likely cause of the characteristic symptoms and signs of the patient's drug reaction?

- A. Anaphylaxis
- B. Non-febrile elevation of body temperature
- C. Non-immunogenic histamine release
- D. Prostaglandin D2 release
- E. Sympathetic nervous system stimulation

Hint: Beta-lactams Notes Handout p57

13. Which one of the following best represents the currently recommended vancomycin dosing strategy for the treatment of serious MRSA infections? (more than one answer)

- A. Clinical efficacy correlates with high peak dose to MIC ratio that produces a greater microbial kill rate.
- B. Clinical efficacy correlates with the total amount of drug in the body determined by the AUC/MIC.
- C. Clinical efficacy correlates with drug concentrations above the MIC for the large majority of the time.
- D. Clinical efficacy correlates with high doses that shift the IC50 on the dose-response curve to the right.

Hint: Notes Handout Beta-lactams p54

14. i) What is vancomycin's antibacterial target? (What does vancomycin bind to?)

ii) What step in bacterial cell wall synthesis is inhibited?

- A. Formation of short peptide interbridges
- B. Inhibition the bactoprenol carrier molecule
- C. Lipidation of lipopolysaccharide membrane
- D. Polymerization of peptidoglycan monomers

Hints: Beta-lactams Notes Handout p55

15. From a Big Picture perspective, what is a way to think about the generations of cephalosporins that might make them easier to remember?

Hint: Think about the drugs in each generation according to their spectrums of activity against aerobic and facultative anaerobic gram-positive and gram-negative bacteria – in the ideal (nonexistent) world where there is no resistance.

	Gram-positive	Gram-negative
1 st -gen.		
2 nd -gen.		
3 rd -gen.		
4 th -gen.		
Advanced-gen.		

16. A 43-year-old woman is being prepared for a laparoscopic hysterectomy. To reduce the risk of surgical wound infection from mixed enteric gram-negative bacilli, including the obligate anaerobic gram-negative rod, *Bacteroides fragilis*, she is given one prophylactic I.V. infusion of an antibiotic 30 minutes prior to the surgery. In what subclass is this antibiotic categorized?

- A. Aminopenicillins
- B. Carbapenems
- C. Cell membrane-active drugs
- D. Cephamycins
- E. Glycopeptides

Information to help you deduce the correct option:

Requirements of antimicrobials used for prophylaxis: The selected agent should be:

- bactericidal,
- nontoxic,
- inexpensive, and
- should have in vitro activity against common organisms that cause wound infection after a specific surgical procedure.

Broad spectrum agents should be avoided.

Hint: Summary Table on p47 of Beta-lactams Notes Handout

17. A 22-year-old sexually active male presents to the ED with complaint of painful urination and presence of a profuse yellow discharge at the urethral meatus of 4 days duration. No known drug allergies (NKDA). Gram-stain of urethral exudate shows gram-negative diplococci in many polymorphonuclear (PMN) leukocytes. The patient is given a diagnosis of uncomplicated gonococcal urethritis (gonorrhea). The standard of care for the treatment this infection is administered. What is it and what dose?

- A. Amoxicillin
- B. Aztreonam
- C. Ceftriaxone
- D. Cefuroxime
- E. Ciprofloxacin
- F. Doxycycline
- G. Penicillin G benzathine

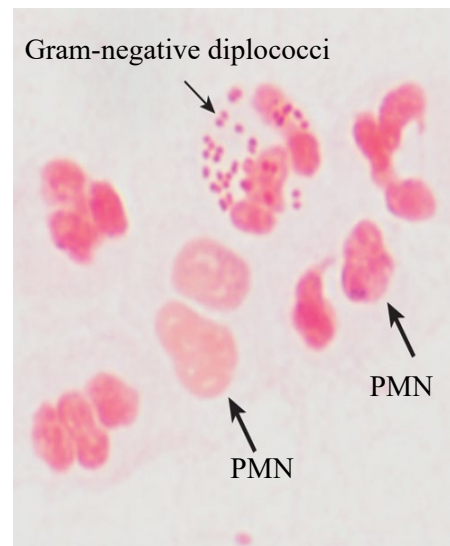
I believe the dosing recommendations for uncomplicated gonorrhea will be useful knowledge in your clinical rotations.

18. Hauser's *Antibiotic Basics for Clinicians* lists the "Six Ps" mnemonic of bacterial resistance mechanisms to the beta-lactam antibiotics: Penicillinase, PBPs, Porins, Pumps, Penetration, and Peptidoglycan. Explain each mechanism.

Hint: Beta-lactams Notes Handout p8.

19. Mr. Nelson is a 60-year-old man who presents to the PCP with a painful, erythematous, nonpurulent infection on his right lower extremity. It began as a small "sore" 3 days ago and rapidly progressed. Possible penicillin allergy manifested by a generalized urticarial rash on his upper body, but no angioedema or shortness of breath, when he took penicillin VK for a "throat infection" as a teenager.

All vital signs are within normal range. Exam reveals an edematous lesion with ill-defined erythema that is warm to the touch and streaky lymphangitis extending proximally. The patient is given a clinical diagnosis of cellulitis and an e-prescription is submitted for an oral antibiotic effective against the presumed infective pathogens, beta-hemolytic streptococci (*S. pyogenes*) and *Staphylococcus aureus*.



Source: S. Riedel, J.A. Hobden, S. Miller, S.A. Morse, T.A. Mietzner, B. Detrick, T.G. Mitchell, J.A. Sakanari, P. Hotez, R. Mejia Jawetz, Melnick, & Adelberg's Medical Microbiology, 28e Copyright © McGraw-Hill Education. All rights reserved.



Source: S. Kang, M. Amagai, A.L. Bruckner, A.H. Enk, D.J. Margolis, A.J. McMichael, J.S. Orringer: Fitzpatrick's Dermatology, Ninth Edition Copyright © McGraw-Hill Education. All rights reserved.

Cellulitis with ill-defined erythema on the lower leg. Note the subtle, streaky lymphangitis extending proximally.



Citation: Chapter 151 Cellulitis and Erysipelas, Kang M, Amagai M, Bruckner AL, Enk AH, Margolis DJ, McMichael AJ, Orringer JS, Fitzpatrick's Dermatology, 9e2019, Figure 151-1. Available at: <https://accessmedicine.mhmedical.com/content.aspx?sectionid=21043491&bookid=251708&resultid=2> Accessed: 06/06/2021 Copyright © 2021 McGraw-Hill Education. All rights reserved.

Which of the following options best represents this approach to therapy?

- A. Definitive therapy
- B. Empiric therapy
- C. Preemptive therapy
- D. Prophylactic therapy
- E. Suppressive therapy

20. Mr. Nelson's skin testing is positive for penicillin allergy and negative for other antibiotics. His mild non-purulent cellulitis is treated empirically with an oral cell wall synthesis inhibitor that has reliable activity against the potential infectious microbes, *Streptococcus* spp and penicillinase-producing *S. aureus*, and has a relatively narrow spectrum of activity. It is unlikely to trigger an allergic reaction in this patient.

The patient has no risk factors for MRSA.

i) Which one of the following best fits this description of the selected drug?

- A. Dicloxacillin
- B. A first-generation cephalosporin
- C. A second-generation cephalosporin
- D. A third-generation cephalosporin
- E. A glycopeptide

ii) What is the name of the selected drug?

21. A usually healthy, active 17-year-old female presents to the pediatrician with complaint of a sore, scratchy throat, productive cough, stuffy, runny nose with thick, yellowish mucus, sneezing, headache, and general ill-feeling (malaise). The symptoms came on yesterday. Ibuprofen helped relieve the headache. NKDA.

Significant findings are T 99.9°F; pharyngeal erythema, nasal mucosal swelling and congestion; lungs are clear to auscultation. The patient is advised on the management of her illness.

With antibiotic stewardship practices in mind, which one of the following recommendations would be the most appropriate course of action for this patient?

- A. Prescribe empiric amoxicillin
- B. Prescribe empiric anti-influenza medication
- C. Recommend ENT consult
- D. Recommend symptomatic treatment

22. Amoxicillin's pharmacokinetic properties are generally typical of the beta-lactams class, in addition to some unique features.

Which choice, A or B, best describes amoxicillin's PK properties?

A Oral Excellent absorption Excellent, reliable blood levels Widely distributed Low protein binding Not metabolized Excreted in urine as unchanged drug 2 to 3 times daily dosing	B Oral Bioavailability 50% to 75%, reduced by food Widely distributed Protein binding >95% Some hepatic metabolism, biliary excretion Excreted in feces / urine as unchanged and metabolites Dose adjustment in renal impairment not usually required. 4 times daily dosing
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ii) Which drug has PK properties represented in the other choice?

iii) Reflection: What is the clinical relevance of the drugs' pharmacokinetic profiles?

23. A 2-year-old boy is brought to the pediatrician's office by his mother. He has been tugging on his ear since yesterday after his morning nap and has been running a temperature of 101.3F (38.5C) taken by ear thermometer at home. He has been irritable; his appetite and fluid intake have been good. A bulging, white tympanic membrane is visualized on otoscopy. The patient has had two bouts of acute otitis media treated with antibiotics, 3 months ago and 4 weeks ago, respectively. He is given a clinical diagnosis of recurrent acute otitis media. An e-prescription is submitted for an antibiotic at a high dose intended to eradicate bacterial infection by *H. influenzae*, *M. catarrhalis*, and *Streptococcus pneumoniae* strains expressing intermediate resistance. Which one of the following statements best defines the profile of the gram-positive bacterial strain with the resistance pattern described in the vignette?

- A. IC50 shifts to the right; Emax is achieved.
- B. IC50 shifts to the right; the response curve is depressed.
- C. High peak concentration to MIC ratio (C_{max}/MIC) will achieve a greater rate of bacterial killing.
- D. High dose increases the drug's half-life for improved efficacy.

24. A 64-year-old man with a history of advanced hepatic cirrhosis and ascites presents to the emergency department with severe abdominal pain. He has had multiple visits to the hospital in the past few months, including multiple visits for ascites drainage and antibiotics. He has been taking fluoroquinolone antibiotic prophylactic therapy. His abdomen is distended with significant tenderness on palpation. Upon further evaluation, he is diagnosed with spontaneous bacterial peritonitis (SBP) severe disease. Ascitic fluid, blood, and urine samples are obtained for identification and susceptibilities; ascitic fluid is sent for PMN (polymorphonuclear leukocytes) count.

i) Which of the following is the best course of action?

- A. Hold treatment while waiting for culture and sensitivity results.
- B. Run DNA-based PCR testing for microbial resistance.
- C. Initiate empiric broad-spectrum antibiotic therapy.
- D. Start the standard initial drug regimen for BPS (non-critically ill patients).

ii) What is the next step regarding antibiotic therapy that should be included in the treatment plan?

25. A 19-year-old male presents to the ED with complaint of a lesion on his penis. He admits to having sex with a female sex worker about 1 month ago. He reports that he has never done anything like it before and has not had a sexual encounter since that time. He is otherwise healthy. He does not use recreational drugs, alcohol, or nicotine-containing products. NKDA. Physical findings: a painless, non-suppurative chancre the penis, firm bilateral lymphadenopathy. All other findings are unremarkable. Serological testing confirms *Treponema pallidum* infection. Screening for HIV and other STIs is negative. He is given the standard treatment, which induces rapid resolution and long-acting effect. A follow-up appointment is scheduled with the hospital's family practice clinic.

What is the preferred drug for this patient's early-stage disease?

- A. Amoxicillin 3 gm plus probenecid 500 mg both orally twice daily x14 days
- B. Azithromycin 2 gm po x1 dose
- C. Ceftriaxone 1 gm I.M. daily x10-14 days
- D. Doxycycline 100 mg orally every 12 hours x 14 days
- E. Penicillin G benzathine 2.4 million units, I.M. x1 dose
- F. Tetracycline 500 mg orally four times daily x14 days

26. A 2-week-old male infant is brought to the ED by his parents because of fever, irritability, lethargy, and poor tone for 2 hours. The patient is ultimately diagnosed with bacterial meningitis and admitted to the NICU for treatment. The combination antibiotic regimen includes cefotaxime. Another 3rd-generation cephalosporin with the same spectrum of activity is not used in this patient because of potential adverse effects: 1) displacement of bilirubin on plasma proteins, increasing the risk of kernicterus (encephalopathy), and 2) potentially fatal lung and kidney damage due to drug-calcium precipitates when calcium-containing fluids are coadministered.

i) Which one of the following is the described drug that should be avoided in this patient?

- A. Cefazolin
- B. Cefotaxime
- C. Ceftazidime
- D. Ceftriaxone
- E. Cefepime

ii) Which is the preferred drug therapy for this patient?

- A. Cefazolin
- B. Cefotaxime
- C. Ceftazidime
- D. Ceftriaxone
- E. Cefepime

27. What properties of ceftriaxone and cefotaxime are the same and what properties are different?

Why is it important to think of these two drugs together?

CHALLENGE: Write down on paper as much as you can from memory. Then check. Then write it down from memory again.

Hint: Beta-lactams Notes Handout, p40

28. Empiric antibiotic therapy for meningitis in the premature neonate should be a bactericidal drug having coverage against Group B *Streptococcus*, *E. coli*, and *Listeria monocytogenes*.

i) Which one of the following is included for the eradication of the presumptive *Listeria* infection?

- A. Ampicillin
- B. Cefazolin
- C. Ceftazidime
- D. Cefotaxime
- E. Piperacillin-tazobactam

ii) Why is a bactericidal antibiotic preferred in the treatment of meningitis?

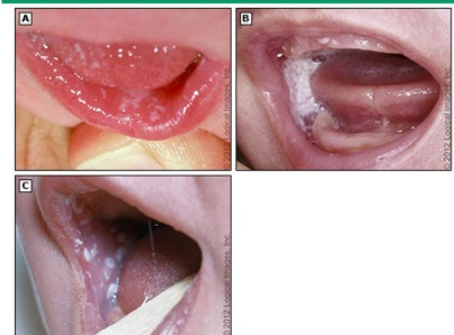
29. A normally healthy, active 4-month-old female presents to the pediatrician with a white coating on her tongue and a diaper rash for 2 days. She has also been having loose bowel movements for one week. Medical history is pertinent for recurrent acute otitis media treated with oral amoxicillin-clavulanate for the past 8 days of a 10-days course of therapy.

Otoscopy shows that the acute otitis media has resolved. White plaques are noted on the buccal mucosa, palate, tongue, and oropharynx. Confluent erythema with erythematous papules and plaques is noted in the inguinal area. Findings are consistent with a drug-induced secondary infection. The exam is otherwise unremarkable.

The child's mother is directed to continue the antibiotic until it is finished. She is given instructions on the use of a medication for management of the oropharyngeal infection and a topical cream for the diaper dermatitis. What secondary infection are these medications intended to treat?

Short answer

Three examples of thrush (oral candidiasis) in an infant



Note the white plaques on the inner lip and tongue (panel A) and on the buccal mucosa (panels B and C).

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Candidal diaper dermatitis



Erythematous plaques and satellite pustules in an infant.

Courtesy of Anthony Mancini, MD, Children's Memorial Hospital, Chicago, IL.

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30. An adult with recurrent rhinosinusitis with indication for antibiotic therapy is given amoxicillin-clavulanate for coverage of ESBL-producing *Haemophilus influenzae* and *Moraxella catarrhalis*.

Which one of the following is a common classification of these beta-lactamases?

- A. Ambler Class A serine beta-lactamases
- B. Ambler Class B metallo-beta-lactamases
- C. Ambler Class C serine beta-lactamases
- D. Ambler Class D serine beta-lactamases

What does ESBL stand for?

31. A 78-year-old obese woman with a long history of type 2 diabetes is hospitalized for treatment of an acute complicated urinary tract infection. She resides in a nursing home and has been hospitalized three times in the past six months for the treatment of recurrent urinary tract infections. She is given intravenous infusions of meropenem-vaborbactam.

What bacterial enzymes are inhibited by the beta-lactamase inhibitor paired with the antibiotic in the patient's therapy?

- A. Class A ESBLs
- B. Class A KPCs
- C. Class B metallo-beta-lactamases
- D. Class C Amp C beta-lactamases
- E. Class D OXA beta-lactamases

32. A 40-year-old woman is treated for a skin abscess with debridement and an oral drug with activity against streptococci and MRSA. Culture and sensitivity results show that the infection is caused by methicillin-sensitive *Staphylococcus aureus* (MSSA). The antibiotic regimen is changed to cephalexin. This course of action is referred to as:

- A. Definitive therapy
- B. Preemptive therapy
- C. Presumptive therapy
- D. Suppressive therapy

33. A 14-year-old female cystic fibrosis patient takes an inhaled monobactam antibiotic via nebulizer for the treatment of her chronic lung disease. The drug is being used for the management of infection caused by an obligate aerobic gram-negative, non-fermenting, oxidase-positive rod commonly present in moist environments. This organism colonizes mucous membranes or skin and invades locally. It can cause serious, potentially life-threatening infections in immunocompromised and medically vulnerable patients.

The selected drug has activity against aerobic gram-negative organisms only.

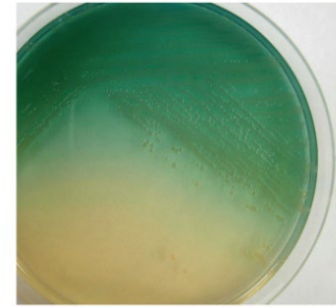
What is the selected antibiotic?

- A. Aztreonam
- B. Cefazolin
- C. Daptomycin
- D. Ertapenem

Hint: Students, *you can come up with the correct answer from the last statement*. You do not need to know the microbiology of the organism.

Blue-green pigment visible on nutrient agar is unique to *P. aeruginosa*.

Notes p51



Source: S. Riedel, J.A. Hobden, S. Miller, S.A. Morse, T.A. Mietzner, B. Detrick, T.G. Mitchell, J.A. Saksena, P. Hotez, R. Muga Javetz, M. S. Adelman's Medical Microbiology, 20e Copyright © McGraw-Hill Education. All rights reserved.

P. aeruginosa on a 10-cm Mueller-Hinton agar plate. Individual colonies are 3–4 mm in diameter. The organism produces pyocyanin, which is blue, and pyoverdine, which is green. Together these pigments produce the blue-green color that is seen in the agar around the *Pseudomonas* growth. (Courtesy of S. Lowe.)

Citation: Chapter 16 Pseudomonas, Acinetobacter, Burkholderia, and Brevibacterium, Riedel S, Hobden JA, Miller S, Morse SA, Mietzner TA, Detrick B, Mitchell TG, Saksena JA, Hotez P, Muga R, Javetz R, Adelman's Medical Microbiology, 20e, 2019 Available at: <https://accessmedicine.mhmedical.com/content.aspx?sectionid=21777113&bookid=242987&edition=2> Accessed October 31, 2020 Copyright © 2020 McGraw-Hill Education. All rights reserved.

34. An 84-year-old mechanically ventilated male with a long history of chronic obstructive pulmonary disease develops new-onset fever, purulent sputum, leukocytosis, and sustained respiratory deterioration. Microscopic examination of bronchoscopic specimens shows gram-negative rods. Culture and sensitivity results show carbapenem-resistant Enterobacterales (CREs). The patient is given IV infusion with a siderophore antibiotic effective against multidrug resistant GNBs including some Ambler class D OXA-type beta-lactamases.

- i) What is the unique property of this drug?
- ii) What is the mechanism of antibacterial action of this drug?
- iii) What is the name of the drug?

35. Emily is a D.O. who starts her own family practice clinic in a rural area. She recognizes the need for a coordinated program to achieve optimal outcomes for the clinic's patients. One of her focus areas is the appropriate use of antimicrobial agents.

- She communicates with her clinical and office staff about best practices for antimicrobial use, including education of the patients and caregivers.

- She provides her clinical staff with current references and guidelines for effective antimicrobial selection.
- The office manager is assigned the tasks of displaying patient information pamphlets describing appropriate antibiotic use and printing daily reports on antibiotic prescriptions obtained from the EMR.
- Emily provides regular feedback to the individual prescribers and progress reports to the clinical and administrative staff.

What is this systematic measurement and coordinated interventions to promote the optimal use of antimicrobials is known as?

Short answer

36. Matching: Sometimes it is easier to remember the few drugs active against a certain microbe than memorizing the many that are not.

P. aeruginosa can cause serious and potentially life-threatening infections in immunocompromised and medically vulnerable patients.

P. aeruginosa is intrinsically resistant to many antibiotics, can form biofilms on tissues and medical devices, produces enzymes and toxins damaging to host tissues, and can lead to sepsis.

Highlight the beta-lactams that have activity to susceptible *P. aeruginosa*. (In other words, *P. aeruginosa* is not intrinsically resistant to them.)

(Only commonly used parenteral agents are listed.)

Penicillins	Cephalosporins		Carbapenems	Monobactam
Penicillin G	1 st -gen	Cefazolin	Imipenem-cilastatin	Aztreonam
Ampicillin	2 nd -gen	Cefoxitin	Meropenem	
Piperacillin-tazobactam	3 rd -gen	Ceftazidime	Ertapenem	
	3 rd -gen	Ceftriaxone		
	4 th -gen	Cefepime		
	Adv-gen	Ceftaroline		
	Adv-gen	Ceftolozane-tazobactam		
	Novel	Cefiderocol		